



Individual Documentation Of

Disease Detection_AI-Chatbot

Name : Jahin Nowaz Sharoj

ID : 0112230482

Course : Artificial Intelligence Lab

Section : D

Team Name : Sentinel

1) Problem Description :

For the medical field, diagnosing the disease properly based on the patient's described symptoms is a much difficult task to do. For most of the diseases, the symptoms are quite similar to another, that's why it is a complex activity to detect the actual disease of the patient.

Our project's main idea is about the solution of the following problem. The core idea of our project is to develop an automated, intelligent based Disease Detection and prediction system.

By giving a dynamic list of the symptoms, we want to train our data properly with different machine learning models. Then after training, our built system will be able to detect the disease according to the symptoms that the user will be able to select in the web-application.

The system aims to serve as a preliminary diagnostic tool to connect the difference between initial symptom onset and professional medical advising.

2) Methodology :

Our project will be followed by a Supervised Machine Learning approach.

- **Model Selection** : Support vector machine is chosen as one of the models that we will use. I have implemented the SVM, other members will do other models. SVM is a highly effective ML model for a large number of datasets. As our dataset is quite large, SVM will be a perfect fit to get a much better result after training our dataset. A linear Kernel was used for the categorical data mapping to numerical weights that often follows linear separability.

- **Data Transformation** : Our methodology involved a data-mapping strategy where each string symptom was assigned a specific numerical weight based on a reference table.
- **Evaluation Strategy** : For evaluating the model fairly across many diseases, the dataset was split into 80% training purpose and 20% for testing to ensure equal accuracy of the detection of each disease. Model accuracy was measured by calculating Accuracy, Precision, Recall and F-1 score with macro-averaging to ensure the rare diseases were weighted equally to common ones.

3) Implementation Details :

- **Data Preprocessing** : Data cleaning was strictly maintained to handle inconsistencies in the raw CSV file data. Trailing spaces were stripped, and spaces between words were replaced with underscore to create uniform keys. Missing symptom columns in the dataset were assigned with 0. The textual symptoms were mapped to their integer weights , creating a numerical feature matrix (X) that mapped to their target disease (Y).
- **Backend & Model** : Python backend was built using the Flask web framework. Upon initialization , the server pre-loads the datasets, preprocesses the arrays, and trains the Support Vector Classifier.

4) Results & Analysis :

Accuracy : 97.26 %
Precision(Macro) : 97.39 %
Recall (Macro) : 97.26 %
F1-Score (Macro): 97.28 %

These results indicate that the use of correlation of symptoms with severity weights produces extremely unique spaces for the classification of the SVM. The use of an F1-Score above 97% is highly commendable given that it is a multiclass problem with over 40 labels in all. The visual interpretation of the Confusion Matrix proved that the classifier was quite efficient in predicting most disease classifications apart from the minor errors that could occur due to similarities in the symptoms of such diseases (cold vs. allergies). Ultimately, the linear SVM proved to be extremely accurate and flexible.